

Brandon E Martinez

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EDUCATION

University of Notre Dame

Bachelor of Science in Computer Science

Notre Dame, IN

Expected May 2027

- Relevant Coursework: Data Structures, Systems Programming, Operating System Principles, Computer Architecture, Linear Algebra and Differential Equations, Intro to Artificial Intelligence, Microeconomics

EXPERIENCE

Software Engineer Intern

June 2026 – August 2026

Bank of America

Pennington, NJ

- Built a release-readiness PR analysis service as a **FastAPI** backend with a custom **Bitbucket** REST wrapper and a **React** diff viewer, flagging deployment-blocking DML/DDL scripts, exposed secrets, and breaking API changes before release-branch merges
- Won **1st place of 22 teams** at an internal codeathon with that service, which a senior developer sponsored and adopted for continued development
- Integrated **Apache Kafka** as the transport for an incident-creation pipeline and authored the **Spring** JDBC data-access layer behind it
- Built parser and connection classes in Spring that automated incident creation through **ServiceNow**, replacing a manual intake step

Undergraduate Software Lead

June 2024 – Present

Domer Rover

Notre Dame, IN

- Led the software subteam through a ground-up rework of the autonomy stack, raising the team's official University Rover Challenge software score **15%** and autonomous navigation from **6.2 to 8.0**
- Wrote a **670-line C++ ros2_control** hardware interface driving RoboClaw 2x60A controllers over USB-UART, delivering open-loop velocity control for a six-wheel base with no encoders
- Configured a **Nav2** autonomy stack in **ROS2** with **NavFn** planning, **Regulated Pure Pursuit** control, **slam_toolbox** mapping, and layered costmaps with recovery behaviors
- Containerized the development environment in headless and noVNC **Docker** images, unblocking remote Gazebo simulation and on-site hardware work for the subteam

Incoming Teaching Assistant, Embedded Systems

Fall 2026

University of Notre Dame

Notre Dame, IN

PROJECTS

OS Internals | C, Make, Shell Scripting, Unix/Linux

Jan. 2025 – Dec. 2025

- Re-implemented the **POSIX** heap allocator in **C** with an explicit free list, block splitting and coalescing, and instrumentation counters
- Built a process-queue shell with pluggable **FIFO** and **round-robin** CPU schedulers, and a multithreaded message-queue client speaking HTTP over raw **sockets**

CozyCubes | Next.js, TypeScript, IndexedDB, Docker, GitHub Actions

Feb. 2026 – Present

- Built a customizable speedcubing timer in **Next.js** with **WCA**-compliant scramble generation and **IndexedDB** persistence, delivering full session history with no accounts
- Automated deployment to Vercel through a **GitHub Actions** CI/CD pipeline with Docker-containerized self-hosted test environments

Customer Sentiment Analysis | Python, Scikit-learn, Pandas, NumPy, Jupyter

Sep. 2024 – Dec. 2024

- Built sentiment classification models over **17,000+** product reviews with **VADER** and **RoBERTa**, reaching **85%** accuracy by correcting severe class skew with **SMOTE** oversampling

TECHNICAL SKILLS

Languages: TypeScript, Python, Java, C/C++, SQL, Bash

Frameworks/Libraries: React, Next.js, Node.js, FastAPI, Spring, ROS2, PyTorch, Scikit-learn

Developer Tools: Git, Docker, GitHub Actions, Unix/Linux